2-amaliy mashg'ulot.

Ikkinchi va uchinchi tartibli determinantlar va ularni hisoblash. O'rin almashtirishlar. $n-$tartibli determinantning ta'rifi.

Ikkinchi tartibli determinant deb $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$ (2.1)

tenglik bilan aniqlanadigan songa aytiladi. Qisqacha, Δ deb belgilanadi. Bu yerda $a_{11}, a_{12}, a_{21}, a_{22}$ -determinantning elementlari deyiladi. .

a_{11}, a_{12} va a_{21}, a_{22} mos ravishda determinantning 1- va 2-satrlari, a_{11}, a_{21} va a_{12}, a_{22} mos ravishda determinantning 1- va 2-ustunlari deyiladi. Ya'ni

$$a_{ij} : \begin{cases} i - \text{satr tartibi} \\ j - \text{ustun tartibi.} \end{cases}$$

Determinantning ixtiyoriy satri yoki ustuni determinantning *qatori* deb ataladi. a_{11}, a_{22}-elementlar joylashgan diagonal *bosh diagonal* deyiladi. a_{21}, a_{12}-elementlar joylashgan diagonal *yordamchi diagonal* deyiladi.

1-misol. Hisoblang: $\begin{vmatrix} 1 & -22 \\ 7 & 8 \end{vmatrix}$.

► (2.1) formulani qo`llaymiz:

$$\begin{vmatrix} 1 & -22 \\ 7 & 8 \end{vmatrix} = 1 \cdot 8 - (-22) \cdot 7 = 8 + 154 = 162. \blacktriangleleft$$

2-misol. Hisoblang: $\begin{vmatrix} tgx & \sin x \\ \sin x & \operatorname{ctg} x \end{vmatrix}$.

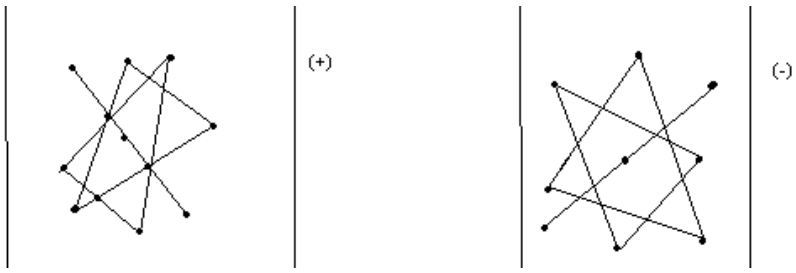
► $\begin{vmatrix} tgx & \sin x \\ \sin x & \operatorname{ctg} x \end{vmatrix} = tgx \cdot ctgx - \sin x \cdot \sin x = 1 - \sin^2 x = \cos^2 x$. ◄

Uchinchi tartibli determinant deb

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32} \quad (2.2)$$

tenglik bilan aniqlanadigan songa aytiladi. Ko`pincha, determinant tartibiga mos ravishda Δ_3 deb ham belgilanadi.

Uchinchi tartibli determinant uchta satr(gorizontal qato (vertikal qator) va to'qqizta elementlarga ega. Yulduzcha sxemasic

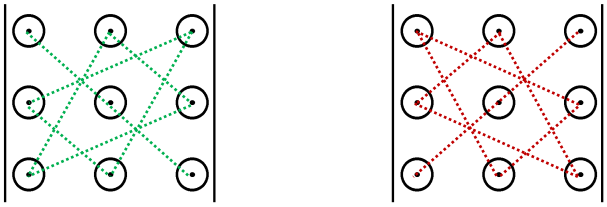


3-misol. $\begin{vmatrix} 1 & 2 & 3 \\ -1 & 3 & 4 \\ 2 & 5 & 2 \end{vmatrix}$ hisoblansin.

Yechish. Ta’rifga binoan.

$$\begin{vmatrix} 1 & 2 & 3 \\ -1 & 3 & 4 \\ 2 & 5 & 2 \end{vmatrix} = 1\cdot 3\cdot 2 + 2\cdot 4\cdot 2 + (-1)5\cdot 3 - 3\cdot 3\cdot 2 -$$
$$-1\cdot 5\cdot 4 - 2(-1)\cdot 2 = 6 + 16 - 15 - 18 - 20 + 4 = -27.$$

Determinant hisoblashning (2.2) formulasini eslab qolish uchun quyidagi sxemani keltiramiz:



Hisoblashning bu qoidasi *uchburchak usuli*(*Sarryus usuli*) deyiladi. Qulaylik uchun deteminantning birinchi va ikkinchi ustunini quyidagicha parallel ko`chirib, bosh diagonal va ikkinchi diagonalga parallel chiziqlar bo`yicha ko`paytmalar tuzamiz:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \begin{matrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{matrix} - \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \begin{matrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{matrix} =$$

$$= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32} \qquad (2.3)$$

bunda bosh diagonal bo`yicha hosil qilinga qo`shuluvchilar musbat ikkinchi diagonal bo`yicha hosil qilingan qo`shiluvchilar manfiy olinadi.

4-misol. $\begin{vmatrix} 1 & 1 & 4 \\ -1 & 2 & 3 \\ -3 & 2 & 5 \end{vmatrix}$ determinantni hisoblang.

► Determinantning birinchi va ikkinchi ustunini parallel ko`chirib yozib Sarryus usulida hisoblaymiz:

$$\begin{vmatrix} 1 & 1 & 4 \\ -1 & 2 & 3 \\ -3 & 2 & 5 \end{vmatrix}\begin{matrix} 1 & 1 \\ -1 & 2 \\ -3 & 2 \end{matrix} = 1\cdot 2\cdot 5 + 1\cdot 3\cdot(-3) + 4\cdot(-1)\cdot 2 - 4\cdot 2\cdot(-3) - 1\cdot 3\cdot 2 - 1\cdot(-1)\cdot 5 =$$

$$= 10 - 9 - 8 + 24 - 6 + 5 = 16. ◄$$

5. Misol. Tenglamani yeching

$$\begin{vmatrix} x^3 & -2 \\ x^2 & 1 \end{vmatrix} = \begin{vmatrix} x & -2 \\ 1 & 1 \end{vmatrix}$$

Yechish. $x^3 + 2x^2 = x + 2;$ $\quad (x+2)(x^2 - 1) = 0;$

J.: $\{-1; 1; -2\}$

Auditoriya topshiriqlari

1. Berilgan ikkinchi tartibli deteminantlarni hisoblang.

a) $\begin{vmatrix} 5 & 3 \\ 7 & -4 \end{vmatrix}$; b) $\begin{vmatrix} 4 & -7 \\ -2 & -3 \end{vmatrix}$; d) $\begin{vmatrix} tgx & -1 \\ 1 & tgx \end{vmatrix}$.

2. Tenglamani yeching.

a) $\begin{vmatrix} x+3 & 2 \\ 7 & x-2 \end{vmatrix} = 0$; b) $\begin{vmatrix} \sin 2x & -\cos 2x \\ \sin 3x & \cos 3x \end{vmatrix} = 0$.

3. Berilgan uchinchi tartibli deteminantlarni hisoblang.

a) $\begin{vmatrix} 1 & -2 & 4 \\ -3 & 5 & 5 \\ 2 & -1 & 3 \end{vmatrix}$; b) $\begin{vmatrix} 10 & -2 & 4 \\ -15 & 3 & 6 \\ 20 & -1 & 5 \end{vmatrix}$; d) $\begin{vmatrix} 1 & 2 & 5 \\ 5 & -3 & 7 \\ 4 & 6 & 5 \end{vmatrix}$.

4. Berilgan uchinchi tartibli deteminantlarni satr yoki us yoyib hisoblang.

a) $\begin{vmatrix} 5 & 0 & 6 \\ 4 & 0 & 5 \\ 2 & 4 & 3 \end{vmatrix}$; b) $\begin{vmatrix} 2 & 2 & -1 \\ 7 & 0 & 3 \\ 3 & 4 & 0 \end{vmatrix}$, d) $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{vmatrix}$.

5. Berilgan deteminantlarni uchburchak shakliga keltirib hisoblang.

a) $\begin{vmatrix} 2 & -3 & 2 & 4 \\ -3 & 2 & 2 & 5 \\ 1 & 5 & -3 & 0 \\ 0 & -1 & 1 & 2 \end{vmatrix}$; b) $\begin{vmatrix} 4 & 0 & -3 & 5 \\ 3 & 2 & -2 & 1 \\ 1 & 3 & 1 & 0 \\ 5 & 6 & 2 & -1 \end{vmatrix}$.

6. $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$ deteminantlarni $a-b$, $a-c$ va $b-c$ larga bo`linishini isbotlang.

7. $\begin{vmatrix} 1 & 6 & 9 \\ 2 & 3 & 4 \\ 3 & 2 & 5 \end{vmatrix}$ deteminantni hisoblamasdan, 13 ga bo`linishini isbotlang.

8. **Misol**. Noma'lum X ni shunday tanlangki, quyidagi tengliklarni qanoatlantirsin.

$a)\begin{vmatrix} x & x & x \\ 2 & -1 & 0 \\ 7 & 4 & 5 \end{vmatrix}=0;\quad b)\begin{vmatrix} 1 & x & x^2 \\ 1 & 2 & 4 \\ 1 & -1 & 1 \end{vmatrix}=0;\quad c)\begin{vmatrix} 4 & x & x \\ x & 4 & x \\ x & x & 4 \end{vmatrix}=0;\quad d)\begin{vmatrix} x & 4 & 4 \\ 4 & x & 4 \\ 4 & 4 & x \end{vmatrix}=0;$

9. **misol.** Tenglamani yeching

$$\begin{vmatrix} x & x \\ -4 & x+1 \end{vmatrix}=0$$

10. Misol. $|A|=|A^T|$ Tenglikni tekshiring

$$A=\begin{vmatrix} 4 & 1 & 6 \\ 7 & 2 & 9 \\ 3 & 0 & 8 \end{vmatrix}$$

11**. misol.** Tengsizlikni yeching

a) $\begin{vmatrix} x^2+7 & 2 \\ 3x & 1 \end{vmatrix}>-1$ J.: $x\in(-\infty;2)\cup(4;\propto$

b) $\begin{vmatrix} x^2+3 & 2 \\ 2 & 5 \end{vmatrix} > 5$ J.: $x \in (-\infty;\infty)$.

12. Quyidagi determinantlarni hisoblang.(1-6)

1. $\begin{vmatrix} 3 & 2 \\ 4 & 5 \end{vmatrix}$ 2. $\begin{vmatrix} -1 & 6 \\ 0 & 3 \end{vmatrix}$ 3. $\begin{vmatrix} \sqrt{a} & -1 \\ \sqrt{a} & 1 \end{vmatrix}$

4. $\begin{vmatrix} a+b & a-b \\ a+b & a-b \end{vmatrix}$ 5. $\begin{vmatrix} \cos\alpha & \sin\alpha \\ \sin\alpha & \cos\alpha \end{vmatrix}$ 6. $\begin{vmatrix} -2 & 2 \\ a-2 & 1 \end{vmatrix}$

13. Tenglamani yeching.(1-4)

1. $\begin{vmatrix} x & x+1 \\ -4 & x+1 \end{vmatrix} = 0$ 2. $\begin{vmatrix} \cos 8x & -\sin 5x \\ \sin 8x & \cos 5x \end{vmatrix} = 0$

3. $\begin{vmatrix} 3 & x & -x \\ 2 & -1 & 3 \\ x+10 & 1 & \end{vmatrix} = 0$ 4. $\begin{vmatrix} x^2 & 4 & 9 \\ x & 2 & 3 \\ 1 & 1 & 1 \end{vmatrix} = 0$

14. Uchinchi tartibli determinantlarni hisoblang.(1-5)

1. $\begin{vmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{vmatrix}$ 2. $\begin{vmatrix} 3 & 4 & -8 \\ 8 & 7 & -2 \\ 2 & -1 & 8 \end{vmatrix}$ 3. $\begin{vmatrix} a & 1 & a \\ -1 & a & 1 \\ a & -1 & a \end{vmatrix}$

4. $\begin{vmatrix} 2 & -3 & 1 \\ 6 & -6 & 2 \\ 2 & -1 & 2 \end{vmatrix}$ 5. $\begin{vmatrix} a & -a & a \\ a & a & -a \\ a & -a & -a \end{vmatrix}$

15. Quyidagi determinantlarni eng qulay yo'l yoki ustun elemyentlari yoyib hisoblang.(1-7)

1. $\begin{vmatrix} 3 & 2 & 4 \\ 0 & 1 & 3 \\ 2 & 0 & 0 \end{vmatrix}$ 2. $\begin{vmatrix} 2 & 0 & 4 \\ 5 & -2 & 1 \\ 1 & 2 & 3 \end{vmatrix}$ 3. $\begin{vmatrix} 1 & b & 1 \\ 0 & b & 0 \\ b & 0 & -b \end{vmatrix}$

4. $\begin{vmatrix} 2 & -3 & 4 & 1 \\ 4 & -2 & 3 & 2 \\ a & b & c & d \\ 3 & -1 & 4 & 3 \end{vmatrix}$ 5. $\begin{vmatrix} 5 & a & 2 & -1 \\ 4 & b & 4 & -3 \\ 2 & c & 3 & -2 \\ 1 & d & 3 & -4 \end{vmatrix}$ 6. $\begin{vmatrix} 2 & -2 & 1 & 0 \\ 0 & 1 & 2 & -1 \\ 3 & -1 & 2 & 3 \\ 3 & 1 & 6 & 1 \end{vmatrix}$

7. $\begin{vmatrix} 0 & -a & -b & -d \\ a & 0 & -c & -e \\ b & c & 0 & 0 \\ d & e & 0 & 0 \end{vmatrix}$

16. O‘rniga qo‘yishlarni ko‘paytiring.

a. $\varphi_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 1 & 5 & 2 \end{pmatrix}$; $\varphi_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 3 & 1 & 2 & 4 \end{pmatrix}$.

b. $\varphi_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 6 & 4 & 5 & 2 & 1 \end{pmatrix}$; $\varphi_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 1 & 5 & 6 & 3 \end{pmatrix}$.

c. $\varphi_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 4 & 3 & 1 & 2 & 7 & 5 & 6 \end{pmatrix}$; $\varphi_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 3 & 1 & 4 & 2 & 7 & 6 \end{pmatrix}$.

d. $\varphi_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 7 & 5 & 3 & 9 & 1 & 2 & 4 & 6 & 8 \end{pmatrix}$; $\varphi_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 9 & 5 & 7 & 2 & 1 & 3 & 4 & 6 & 8 \end{pmatrix}$

17. Quyidagi shartlar asosida α, β larni toping

a) $(\alpha \quad 6 \quad 7 \quad 1 \quad \beta \quad 5 \quad 3)$ – toq.

b) $(11 \quad 5 \quad 7 \quad \alpha \quad 1 \quad 2 \quad 9 \quad 8 \quad 4 \quad 3 \quad \beta)$ – juft.

18. S_4 dagi barcha toq o‘rniga qo‘yishlarni aniqlang.

19. Qaysi 10ta sondan iboratt o‘rniga qo‘yish eng ko‘p inversiyaga ega? Inversiyalar sonini aniqlang.

20. S_5 dan quyidagi shartlar bo‘yicha o‘rniga qo‘yishlarni aniqlang

1) 4ta inversiyaga ega;
2) 7ta inversiyaga ega;
3) 9ta inversiyaga ega;

4) 11ta inversiyaga ega;

21.O‘rnniga qo‘yishlardagi inversiyalar sonini aniqlang.

a. $(1 \quad 4 \quad 7 \quad \ldots \quad 3n-2 \quad 2 \quad 5 \quad 8 \quad \ldots \quad 3n-1 \quad 3 \quad 6 \quad 9 \quad \ldots \quad 3n)$.

b. $(3 \quad 6 \quad 9 \quad \ldots \quad 3n \quad 2 \quad 5 \quad 8 \quad \ldots \quad 3n-1 \quad 1 \quad 4 \quad 7 \quad \ldots \quad 3n-2)$.

c. $(2 \quad 5 \quad 8 \quad \ldots \quad 3n-1 \quad 3 \quad 6 \quad 9 \quad \ldots \quad 3n \quad 1 \quad 4 \quad 7 \quad \ldots \quad 3n-2)$.

d. $(2 \quad 5 \quad 8 \quad \ldots \quad 3n-1 \quad 1 \quad 5 \quad 7 \quad \ldots \quad 3n-2 \quad 3 \quad 6 \quad 9 \quad \ldots \quad 3n)$.

Mustaqil yechish uchun testlar

1. To`g`ri tengliklarni aniqlang.

1) $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} d & c \\ b & a \end{vmatrix}$, 2) $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & c \\ b & d \end{vmatrix}$, 3) $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} c & d \\ a & b \end{vmatrix}$, 4) $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} b & a \\ d & c \end{vmatrix}$.

A) $1),3)$; B) $1),2)$; D) $2),3)$; E) $3),4)$.

2. $\begin{vmatrix} 2 & 1 & 2 \\ -2 & 3 & 0 \\ 1 & 0 & 2 \end{vmatrix}$ determinantning a_{21}elementining M_{21} minorini toping:

A) 4 B) -4 D) 2 E) -2.

3. $\begin{vmatrix} 2 & 1 & 2 \\ -2 & 3 & 0 \\ 1 & 0 & 2 \end{vmatrix}$ determinantning a_{21}elementining A_{21} algebraik to`ldiruvchisiini toping:

A) 4 B) -4 D) 2 E) -2.

4. $\begin{vmatrix} 0 & 3 & 7 \\ 1 & -3 & 4 \\ 0 & 2 & 6 \end{vmatrix}$ deteminantni hisoblang.

A) 4 B) -4 D) 2 E) -2.

5. Agar n-tartibli determinantning satrlarini teskari tartibda yozib chiqilsa qiymati qanday o`zgaradi?

A) $(-1)^n$ ga ko`payadi; B) $(-1)^{n-1}$ ga ko`payadi; D) $(-1)^{\frac{n(n-1)}{2}}$ ga ko`payadi; E) o`zgarmaydi.

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